

Ethan Duy Minh Dam

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Summary

Data science professional with strong skills in Python, SQL, statistical analysis, and machine learning. Developed and delivered a comprehensive statistics curriculum, increasing student engagement and leveraging hypothesis testing and ANOVA to uncover learning gaps. Experienced in exploratory data analysis and collaborative environments, seeking to apply data-driven experimentation and modeling to enhance user experience.

Education

University of California, San Diego

Bachelor of Science, Data Science

Aug 2025 - Present

La Jolla

- **Coursework:** Principles of Data Science, Programming in Python, Data Structures & Algorithms, Theoretical Foundations of Data Science, Data Science in Practice

Golden West Community College

Associate of Arts, Liberal Arts Computer Science and Technology

Sep 2023 - Aug 2025

Huntington Beach

- **Coursework:** Data Structures, Discrete Structures, Computer Architecture, Linear Algebra, Multivariable Calculus, Physics-Based Calculus, Introduction to Statistics

Work Experience

Golden West College

Statistics Supplemental Instructor

Aug 2024 - Sep 2025

Huntington Beach, CA

- Developed comprehensive curriculum and implemented interactive Python-based Jupyter Notebook demonstrations using pandas and matplotlib to improve engagement and concept retention
- Delivered statistical concept lectures to groups of 30+ students twice weekly, simplifying topics like ANOVA and Hypothesis Testing.
- Analyzed student feedback through exploratory data analysis with Python and pandas to identify learning gaps and refine instructional strategies, fostering an inclusive learning environment
- Managed classroom activities and student behavior to ensure a productive and respectful learning atmosphere, which contributed to a positive and effective educational experience.

Uniqlo

Sales Associate

Oct 2023 - Aug 2024

Costa Mesa, CA

- Provided personalized assistance to a high volume of customers by guiding them through product options and sizing, ensuring a satisfactory shopping experience while maintaining accurate knowledge of current inventory.
- Collaborated with team members to maintain store standards and flow, coordinating floor coverage and promptly addressing customer inquiries during high-traffic shifts, which kept the sales area organized and reduced wait times
- Designed and executed informal A/B tests comparing store display layouts to optimize product placement, improving customer engagement during high-traffic periods

Projects

Global Seismic Gap Analysis | <https://ethandam3.github.io/spark-earthquake/earthquake.html>

Jan 2026 - Present

- Developed a scalable ETL pipeline using PySpark to ingest raw USGS seismic logs, cleaning and transforming unstructured geospatial coordinates into vectorized formats for downstream machine learning tasks.
- Optimized complex time-series transformations using Spark SQL Window Functions, calculating temporal deltas across partitioned fault zones to ensure high-performance querying on historical datasets.
- Utilized Spark MLlib to execute distributed K-Means clustering on seismic data, leveraging Spark's vector assembler to handle feature engineering at scale.
- Processed and structured clustered seismic data to interface with Folium, decoupling the backend aggregation logic from the frontend geospatial visualization.

Analysis for "Friends" TV | <https://github.com/ethandam3/friends-tv-strategy-analysis>

Nov 2025 - Dec 2025

- Conducted an end-to-end analysis of the *Friends* dataset to formulate data-driven production strategies for a potential show reboot.
- Performed bootstrap resampling (5,000 iterations) to evaluate crew performance, identifying director Kevin S. Bright as generating statistically significantly higher-rated episodes than peers.
- Designed an A/B test (permutation test) to analyze plot mechanics, proving that episodes with romantic subplots result in a statistically significant increase in IMDb user ratings.
- Developed custom string parsing algorithms to standardize unstructured director and writer credits, enabling accurate talent attribution across 236 episodes.

Skills

- **Programming Languages:** Python, SQL, C++, Java, Spark
- **Data Analysis & Machine Learning:** Machine Learning, Data Analysis, Data Visualization, Exploratory Data Analysis, Regression Analysis, Hypothesis Testing, ANOVA, Unsupervised Learning, Clustering, K-Means, Geospatial Analysis, A/B Testing
- **Data Tools & Libraries:** NumPy, pandas, Jupyter Notebook, MLlib, Matplotlib, Apache Spark, PySpark, seaborn
- **Interpersonal:** Technical Communication, Public Speaking, Management, Cross-Functional Collaboration, Leadership, Time Management, Stakeholder Management